

ANDREA RAGALZI

BACKEND ENGINEER

Python | REST APIs | System Integrations

Málaga, Spain | EU citizen | Remote in Spain / Hybrid in Málaga

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PROFESSIONAL SUMMARY

Software Engineer specialising in Python backend development, REST APIs and system integrations, with five years of total software development experience across insurance and industrial automation. Comfortable working independently and in international teams, taking software from requirements to production. Currently developing a personal FastAPI application with retrieval-augmented generation (RAG). Seeking Backend Engineer or backend-focused Software Engineer roles, hybrid in Málaga or remote from Spain.

PROFESSIONAL EXPERIENCE

BACKEND DEVELOPER – Insurance Software

Oct 2023 – Feb 2025

HRM – Remote

- Developed Python Lambda functions for a serverless insurance claim automation application on AWS, using Pydantic and hexagonal architecture. Wrote automated tests with pytest and contributed from requirements to production.
- Tested the application with the client in production after release and investigated issues in AWS Step Functions workflows. Contributed to infrastructure and automated deployments.
- Developed new features and resolved defects in an enterprise Java/Spring Boot application within a large international team. Used Jenkins CI/CD pipelines across both projects.

SOFTWARE DEVELOPER – Industrial Automation

Sep 2019 – Nov 2022

Gefit S.p.A. – Alessandria, Italy

- Took responsibility for software development on industrial automation and R&D projects, turning requirements into applications and system integrations.
- Developed Python and Flask applications for operator monitoring and automated data logging, with configurable workflows reused across projects. Designed and created MySQL, SQL Server and PostgreSQL databases.
- Built the software infrastructure and integrations for a YOLOv5-based drum recognition system, collaborating with an engineer responsible for model training.
- Supported deployed applications at customer sites in Italy and the United States, maintained legacy software and introduced Git for version control.

PERSONAL PROJECT

DOCUMENT INTELLIGENT HUB – RAG Application

2025 – Present

Personal project – [Live App](#) | [GitHub](#)

- Developed a FastAPI backend for document upload and processing, using LangChain and ChromaDB for retrieval and source-grounded answers. Added conversation history to support follow-up questions.
- Built a Next.js frontend where users can upload documents, ask questions and view answers with source references. Integrated Firebase authentication with user-scoped access to documents and conversations.

TECHNICAL SKILLS

Backend: Python, FastAPI, Flask, REST APIs, Pydantic

Cloud and deployment: AWS Lambda, AWS Step Functions, Amazon S3, Amazon CloudWatch, Jenkins, Docker

Testing and tools: pytest, JUnit, Git, Linux, Bash

Databases: SQL, MySQL, Microsoft SQL Server, PostgreSQL, MongoDB

Applied RAG: LangChain, ChromaDB, document ingestion, embeddings, vector retrieval, LLM integration

Additional technologies: Java/Spring Boot, Next.js, React, TypeScript, Firebase

EDUCATION

Full Stack Web Developer Bootcamp

Jan 2023 – Jul 2023

Epicode Italia – Remote

Computer Science Coursework

2015 – 2021

Università del Piemonte Orientale

Diploma in Electronics and Electrical Engineering

2015

IIS Ciampini-Boccardo – Novi Ligure, Italy

LANGUAGES

Italian: Native | **English:** B2 – Working proficiency | **Spanish:** B2 – Daily conversational use